In [75]:
```python
import pandas as pd
import numpy as np
```

1. Extraction

In [76]:
```python
# load dataset
df = pd.read_csv("../data/covid_2020.csv")
df.head()
```

Out[76]:

	Row_ID	Accurate_Episode_Date	Case_Reported_Date	Test_Reported_Date	Specimen_Dat
0	9	2019-05-30	2020-05-05	2020-05-05	2020-05-0
1	10	2019-11-20	2020-10-21	2020-11-21	2019-11-2
2	12	2020-01-01	2020-04-24	2020-04-24	2020-04-2
3	13	2020-01-01	2020-05-17	2020-05-17	2020-05-1
4	14	2020-01-10	2020-06-10	2020-06-10	2020-06-0

In [77]:
```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 187795 entries, 0 to 187794
Data columns (total 17 columns):
 #   Column                            Non-Null Count   Dtype  
---  ------                            --------------   -----  
 0   Row_ID                            187795 non-null  int64  
 1   Accurate_Episode_Date             187795 non-null  str    
 2   Case_Reported_Date                187795 non-null  str    
 3   Test_Reported_Date                184082 non-null  str    
 4   Specimen_Date                     186838 non-null  str    
 5   Age_Group                         187795 non-null  str    
 6   Client_Gender                     187795 non-null  str    
 7   Outcome1                          5398 non-null    str    
 8   Reporting_PHU_ID                  187795 non-null  int64  
 9   Reporting_PHU                     187795 non-null  str    
 10  Reporting_PHU_Address             187795 non-null  str    
 11  Reporting_PHU_City                187795 non-null  str    
 12  Reporting_PHU_Postal_Code         187795 non-null  str    
 13  Reporting_PHU_Website             187795 non-null  str    
 14  Reporting_PHU_Latitude            187795 non-null  float64
 15  Reporting_PHU_Longitude           187795 non-null  float64
 16  case_reported_date_column_in_years  187795 non-null  str    
dtypes: float64(2), int64(2), str(13)
memory usage: 24.4 MB
```

In [78]:
```
# check null values
df.isnull().sum()
```

Out[78]:
```
Row_ID                                  0
Accurate_Episode_Date                   0
Case_Reported_Date                      0
Test_Reported_Date                   3713
Specimen_Date                         957
Age_Group                               0
Client_Gender                           0
Outcome1                           182397
Reporting_PHU_ID                        0
Reporting_PHU                           0
Reporting_PHU_Address                   0
Reporting_PHU_City                      0
Reporting_PHU_Postal_Code               0
Reporting_PHU_Website                   0
Reporting_PHU_Latitude                  0
Reporting_PHU_Longitude                 0
case_reported_date_column_in_years      0
dtype: int64
```

2. Transform

In [79]:
```
# standardize column names
df.columns = df.columns.str.lower()
```

In [80]:
```
# handle missing values
df["test_reported_date"] = df["test_reported_date"].fillna("UNKNOWN")
```

```
df["specimen_date"] = df["specimen_date"].fillna("UNKNOWN")
df["outcome1"] = df["outcome1"].fillna("NON-FATAL")
```

In [81]:
```
df.head()
```

Out[81]:

	row_id	accurate_episode_date	case_reported_date	test_reported_date	specimen_date	a
0	9	2019-05-30	2020-05-05	2020-05-05	2020-05-03	
1	10	2019-11-20	2020-10-21	2020-11-21	2019-11-20	
2	12	2020-01-01	2020-04-24	2020-04-24	2020-04-23	
3	13	2020-01-01	2020-05-17	2020-05-17	2020-05-15	
4	14	2020-01-10	2020-06-10	2020-06-10	2020-06-09	

3. Load

In [82]:
```
import os
from dotenv import load_dotenv

load_dotenv()
```

Out[82]: True

In [83]:
```
# read credentials
from urllib.parse import quote_plus
DB_USER = os.getenv("DB_USER")
DB_PASSWORD = quote_plus(os.getenv("DB_PASSWORD"))
DB_HOST = os.getenv("DB_HOST")
DB_PORT = os.getenv("DB_PORT")
DB_NAME = os.getenv("DB_NAME")
```

In [84]:
```
# connect to postgres
from sqlalchemy import create_engine, text
engine = create_engine(f"postgresql+psycopg2://{DB_USER}:{DB_PASSWORD}@{DB_HOST}:{D
```

In [85]:
```
# create schema
with open("../databases/01_schema.sql", "r") as file:
    sql_script = file.read()
```

```python
with engine.begin() as conn:
    conn.execute(text(sql_script))
```

In [86]:
```python
# create PHU dataframe
phu_df = df[["reporting_phu_id",
             "reporting_phu",
             "reporting_phu_address",
             "reporting_phu_city",
             "reporting_phu_postal_code",
             "reporting_phu_website",
             "reporting_phu_latitude",
             "reporting_phu_longitude"]].drop_duplicates()

# create COVID cases dataframe
cases_df = df[["row_id",
               "accurate_episode_date",
               "case_reported_date",
               "test_reported_date",
               "specimen_date",
               "age_group",
               "client_gender",
               "outcome1",
               "reporting_phu_id"]].copy()
```

In [87]:
```python
# load phu table into postgres
phu_df.to_sql("reporting_phu",
          engine,
          if_exists="append",
          index=False,
          chunksize=20000,
          method="multi")
```

Out[87]: 34

In [88]:
```python
# load COVID cases table into postgres
cases_df.to_sql("covid_cases",
          engine,
          if_exists="replace",
          index=False,
          chunksize=20000,
          method="multi")
```

Out[88]: 187795